



WHEELS & CASTORS

PROMAXX INDUSTRIES

**"Built for Performance,
Focused on Value"**





■ About Us



We are specialized in manufacturing wheels and castors for a wide range of trolleys and material handling equipments, ideal for use in factories, warehouses, storage facilities, and other industrial settings. Our product lineup includes a variety of robust castor wheels engineered for diverse applications. Detailed specifications are provided in our catalogue, and our team is available to offer any further information or guidance you may need.

■ Our Vision

“Built for Performance, Focused on Value”



We understand that the real cost of a castor includes downtime and workflow disruptions, so we craft durable, reliable products that keep operations running smoothly. Our commitment to quality has built lasting trust, with most clients returning for dependable solutions.

■ Customer Commitments

- We guarantee 100% on-time delivery.
- We focus on understanding customer needs and offering expert product selection guidance.
- We uphold optimal quality through rigorous monitoring and evaluation.



WHY "PROMAXX" WHEELS & CASTORS?

There are several compelling reasons why numerous satisfied customers choose "PROMAXX" wheels and castors. A few of these reasons include:

Durable Designs, Extended Service Life, Reduced Overall Costs

At Promaxx Industries, our wheels and castors are engineered to be robust while maintaining an attractive appearance. We carefully select raw materials to ensure superior strength and longevity. This significantly lowers the likelihood of failures and minimizes maintenance needs, resulting in fewer operational disruptions and reduced downtime for trolleys and equipment. Although the initial investment may be slightly higher, the prolonged service life and lower maintenance expenses make our products highly cost-effective in the long term.

Extensive Range to Meet Diverse Applications

Promaxx wheels are available in a variety of materials and sizes, from 50 mm to 300 mm in diameter, with load capacities spanning 250 kg to 2000 kg per wheel. We manufacture both fixed and swivel brackets in multiple series to support various load requirements. Our comprehensive selection of wheel types and bracket designs enables customers to choose the ideal configuration that suits their specific application needs perfectly.

Immediate Stock Availability, Wide and Responsive Dealer Network

We ensure high stock levels of all products within our warehouses to guarantee quick dispatch. Furthermore, our extensive network of Authorized Dealers maintains large inventories of essential items, facilitating prompt availability. Our dealers assist customers in selecting appropriate products, offer competitive pricing, ensure timely deliveries, and provide swift after-sales support whenever required.

Commitment to Customer Satisfaction

At Promaxx Industries, we place utmost importance on delivering superior products at fair prices within the shortest possible lead times. We take pride in the fact that approximately 90% of our business comes from repeat customers, a testament to their satisfaction and trust. Once a customer experiences the performance and reliability of Promaxx products, they seldom consider switching to any other brand.

POLYURETHANE WHEELS (PU-CI)

PU-CI wheels are designed by bonding high-abrasion-resistant polyurethane to a cast iron core using a casting process, which enhances the material properties compared to injection-molded alternatives. These wheels deliver excellent load-bearing capacity and extended lifespan, outperforming PU wheels. Additionally, they feature exceptional shock absorption and operate at a low noise level.

- Hardness of the Polyurethane:- (94 ± 5) on the Shore 'A' scale
- Maximum working Temperature:- 50°C .



TECHNICAL DETAILS – POLYURETHANE WHEELS				
Size of wheel D mm * W mm	Ball bearing sizes	Size of bore for wheels with BB d mm	Width of boss of Wheel B mm	Load carrying capacity Kg.
100*50	6205	25.00	60	250
125*50	6205	25.00	60	300
150*50	6205	25.00	60	400
150*75	6205	25.00	90	550
200*50	6205	25.00	60	500
200*75	6205	25.00	90	750
250*50	6205	25.00	60	750
250*75	6205	25.00	90	1000
300*50	6205	25.00	60	900
300*75	6205	25.00	90	1200
TECHNICAL DETAILS – POLYURETHANE ROLLERS				
82*70	6204	20.00	70	300

UHMW-PE WHEELS

Made from high-grade Ultra High Molecular Weight Polyethylene (UHMW-PE), these lightweight wheels offer outstanding load capacity, wear resistance, and impact strength, without damaging floors. The low friction of UHMW-PE reduces pushing or pulling effort, and the material is highly resistant to chemicals and acids. To ensure ball bearing stability, a metal insert with a step bore is added.

HD Type: For higher loads, with a thicker section, designed for HD series casters.



Hardness of the Polyurethane:- (75 ± 5) on the Shore 'D' scale
Maximum working temperature :- 70°C

TECHNICAL DETAILS – UHMW WHEELS						
Size of wheel D mm * W mm	Ball bearing sizes	Size of bore for wheels with			Width of boss of wheel mm	Load carrying capacity Kg.
		PLB d mm	SIB d mm	BB d mm		HD
100*50	6205	--	--	25.00	60	300
125*50	6205	--	--	25.0	60	300
150*50	6205	--	--	25.0	60	550
150*75	6205	25.4	25.4	25.0	90	750
200*50	6205	--	--	25.00	60	750
200*75	6205	25.4	25.4	25.00	90	900
250*50	6205	25.4	25.4	25.00	60	800
250*75	6205	25.4	25.4	25.00	90	1100
300*50	6205	25.4	25.4	25.00	60	1000
300*75	6205	25.4	25.4	25.00	90	1400
TECHNICAL DETAILS – UHMW ROLLERS						
82*70	6204	--	--	20.00	70	400
82*70	6205	--	--	25.00	70	400

CAST IRON WHEELS (CI)

These durable cast iron wheels are manufactured from high-quality, defect-free graded castings, allowing for easy machining and high load capacity. They are ideal for high-temperature applications such as ovens and foundries, although they may cause wear on delicate floor surfaces.

Specifications:

- Casting Hardness: 175 ± 10 BHN
- Casting Grade: FG 200 (IS - 210)
- Max Operating Temperature: 250°C



TECHNICAL DETAILS – CAST IRON WHEELS					
Size of wheel D mm * W mm	Ball bearing sizes	Size of bore for wheels with		Width of boss of Wheel mm	Load carrying capacity Kg.
		SIB d mm	BB d mm		
100*32	6004	15.9	20.00	45	300
100*50	6205	25.4	25.00	60	400
125*50	6205	---	25.00	60	500
150*50	6205	---	25.00	60	700
150*75	6205	---	25.00	90	1050
200*50	6205	---	25.00	60	900
200*75	6205	---	25.00	90	1350
250*50	6205	---	25.00	60	1050
250*75	6205	---	25.00	90	1550
300*50	6205	---	25.00	60	1400
300*75	6205	---	25.00	90	2000

OUR PRODUCTS



UHMW-PE WHEEL



CI WHEEL



PU-CI WHEEL



FIX (HD) 50 WIDTH CASTOR



SWIVEL (HD-C) 50 WIDTH CASTOR



HD-C WHEEL BRAKE



FIX (HD TYPE) 75 WIDTH CASTOR



SWIVEL (HD TYPE) 75 WIDTH CASTOR



HD-F RIVET



FIX (HD TYPE) TWIN CASTOR



SWIVEL (HD TYPE) TWIN CASTOR

DIMENSIONAL DETAILS AND LOAD CARRYING CAPACITIES

FIXED CASTORS (HD) AND SWIVEL CASTORS RIVETTED TYPE (HD-F) SERIES

Size of wheel Dia* Width	Mounting Plate P. C. D	Overall Height	Offset	Load carrying capacity (Kg)		
				PU-CI	UHMW	CI
100*50	60 x 120	155	40	250	300	400
125*50	60 x 120	180	45	300	350	500
150*50	60 x 120	205	50	400	550	700
150*75	60 x 120	205	50	550	750	1050
200*50	60 x 120	255	60	500	750	900
200*75	60 x 120	255	60	750	900	1350
250*50	60 x 120	305	70	650	800	1100
250*75	60 x 120	305	70	1000	1100	1600
300*50	60 x 120	355	80	800	1000	1400
300*75	60 x 120	355	80	1200	1400	1800

SWIVEL CASTORS (HD-C) SERIES

Size of wheel Dia* Width	Mounting Plate P. C. D.	Overall Height	Offset	Load carrying capacity		
				PU-CI	UHMW	CI
100*50	60 x 120	155	40	250	300	400
125*50	60 x 120	180	45	300	350	500
150*50	60 x 120	205	50	400	550	700
200*50	60 x 120	255	60	500	750	900
250*50	60 x 120	305	70	650	800	1100
300*50	60 x 120	355	80	800	1000	1200

HEAVY DUTY FORGED STEEL TWIN CASTORS (HD-TWIN SERIES)

Twin castors offer superior load-bearing performance while maintaining consistent overall height. By distributing the weight across two wheels, they minimize floor wear and enhance stability. Unlike single wide wheels, twin castors reduce scrubbing and rolling resistance during directional changes due to the independent rotation of each wheel. This design ensures smoother mobility, particularly under heavy loads. Optional features such as wheel locks, brakes, and swivel locks further enhance their functionality. In environments where height constraints exist but high load capacity is essential, twin castors present an optimal and efficient solution.



FIX (HD TYPE) TWIN CASTOR



SWIVEL (HD TYPE) TWIN CASTOR

FIXED AND SWIVEL CASTORS (HD-TWIN SERIES)

Size of wheel Dia* Width	Mounting Plate P. C. D	Overall Height	Offset	Load Carrying Capacity (Kg)		
				PU-CI	UHMW	CI
100*50	60 x 120	155	40	500	600	800
125*50	60 x 120	180	45	600	700	1000
150*50	60 x 120	205	50	800	1100	1400
200*50	60 x 120	255	60	1000	1500	1800
250*50	60 x 120	305	70	1400	1600	1800
300*50	60 x 120	355	80	1700	1800	1900

SELECTION CRITERIA FOR WHEELS AND CASTORS

1) Pushing / Pulling Effort

Moving ease depends on wheel size, material, and bearings:

- Diameter of Wheel: Larger diameters reduce effort proportionally; a 100 mm wheel diameter is easier to push than a 50 mm diameter.
- Material: Rubber wheels, with a high friction coefficient, need more effort than cast iron wheels. Order of effort (low to high): cast iron < Nylon < UHMW < PP < PU < rubber.
- Bearings: Effort ranks by bearing type (moderate for PLB & SIB, less for NRB, minimum for BB). Maintenance (regular cleaning and greasing) extends efficiency.

2) Load Carrying Capacity

Improper load assessment leads to around 80% of castor failures. Catalog ratings typically reflect ideal conditions—smooth, level, and non-deformable floors at speeds below 6 km/hr and temperatures around $20 \pm 10^{\circ}\text{C}$ (IS: 7369-1983).

Real-world factors like uneven loads, rough floors, and shock loads can double the strain on individual castors.

- Uneven Loading: Weight imbalances can cause some castors to bear extra load. Assume up to 25% additional load on individual castors.
 - Uneven Floors: On uneven floors, expect a 33% overload on three wheels when one loses contact.
 - Shock Loads: Abrupt jolts, from joints or lifted trolleys, can double the load on the first castor to make contact with the ground.
- *Guideline:* Use only 50-60% of the nominal capacity as a safety buffer.

3) Noise Levels

Noise varies by wheel material:

- Low Noise: Rubber and PU
- Medium Noise: PP, Nylon, UHMW-PE
- High Noise: Cast iron

4) Working Temperature

Wheel load capacity decreases as temperature rises, affecting materials like rubber, PU, PP, Nylon, and UHMW.

Safe Temperature Ranges:

- PU: Up to 50°C
- UHMW-PE: Up to 70°C
- Cast Iron: Up to 250°C

Guideline: At high temperatures, use special greases designed to withstand elevated heat in wheel bearings.

5) Floor Surface

Floor texture affects pushing effort, noise, and wheel life:

- Rough Surfaces: Use UHMW-PE or cast iron wheels with medium to heavy-duty castors. Acceptable noise levels may rise slightly.
- Smooth Surfaces: Rubber, PU, or PP wheels with light-duty castors are optimal.

6) Use with Power-Driven Vehicles

Manual operation at speeds below 6 km/hr is ideal. When using powered movement:

- Recommended Wheels: UHMW-PE or cast iron wheels with heavy-duty brackets
- Load Adjustment: Apply a higher safety factor due to higher shock loads from powered towing.
- The Castors are designed to work for manually pushed or pulled trolleys having speed below 6 km/hr.

Ultimately, the selection of castor wheels represents a balanced compromise based on the number of operating conditions and performance expectations.

Proper selection of wheels and castors is essential to avoid failures.

An optimal solution for applications involving heavy loads that require low-height castors.

Trust our expertise to drive your business growth.



PROMAXX INDUSTRIES



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